

Edward Fields

Conway, AR • 501-697-5863 • edwardspike14@gmail.com • [linkedin.com/in/edward-fields-895b80149](https://www.linkedin.com/in/edward-fields-895b80149) • github.com/efieldsUCA • efieldsuca.github.io



Engineering Physics graduate (May 2026) from the University of Central Arkansas with hands-on experience in electrical design, test automation, and hardware debugging to root cause. As electrical and software lead on an autonomous robot, I designed and built the electrical system, wrote Python test procedures to validate it, and drove failures to root cause on the bench. Won Best Notebook and placed 3rd statewide at the 2026 ASGC (Arkansas Space Grant Consortium). Comfortable across the hardware-software boundary, and a hands-on builder who takes on self-directed projects, including custom PCB design.

EDUCATION

University of Central Arkansas — Conway, AR

May 2026

Bachelor of Engineering Physics (Focus: Robotics & Embedded Systems) • GPA: 3.2 • Presidential Scholar • Dean's List • Alpha Lambda Delta Honor Society

TECHNICAL SKILLS

Test and Automation: Python test automation, test procedure design, validation and characterization, hardware-in-the-loop testing, root-cause analysis

Hardware and Debug: Bench bring-up and debugging with oscilloscopes, function generators, and multimeters, circuit design, power distribution, signal-integrity verification, soldering

Design and Embedded: PCB schematic capture and layout (KiCad), microcontrollers (Raspberry Pi Pico), embedded platforms (Raspberry Pi), serial communication (UART, USB), sensor integration

Software and Tools: Python, ROS2, Git and GitHub, Linux (Ubuntu), C++, FreeCAD

WORK EXPERIENCE

Georg Fischer — R&D Intern

July 2026 – Present

- Develop and execute measurement and test procedures, turning results into documented findings that inform R&D and quality decisions.
- Conduct a dimensional-stability study on manufactured pipes to characterize shrinkage and process behavior.

Walmart — Personal Shopper Conway, AR

May 2022 – Present

- Operate in a high-volume, high-density retail fulfillment environment, sustaining strong accuracy and throughput under time pressure; promoted to lead day-to-day operations (4+ years).

ENGINEERING PROJECTS

Senior Design: WonkyBot

Aug 2025 – May 2026

Electrical & Software Lead — 3rd Place Statewide + Best Notebook — ASGC Competition, Petit Jean, AR

- Designed and built the electrical system, then brought it up on the bench using oscilloscopes, function generators, and multimeters to verify signals and validate behavior.
- Wrote the test procedures in Python that validated the integrated system and built the ROS2 control and navigation stack on a Raspberry Pi 5.
- Drove a failure to root cause, tracing a motor stall to a wiring fault and redesigning the circuit protection.
- Designed the power distribution: LiPo batteries with multi-rail buck converters supplying isolated, protected power to compute, sensors, and drive subsystems.

PERSONAL PROJECTS

- Ciphering Lock (embedded): designed the entire custom PCB myself in KiCad, from schematic capture through layout, and sent it out for fabrication; an embedded randomizing-passcode lock.
- Additional self-directed builds: at-home autonomous robot, automotive electrical work.

PROFESSIONAL DEVELOPMENT

- IEEE Student Showcase 2025 — Presented an autonomous robotics capstone to an IEEE panel; fielded technical questions on system design and testing from industry professionals.
- ASGC Collegiate Networking Events — Attended statewide events at the University of Arkansas (Jan 2026) and hosted at UCA (Feb 2026); connected with engineering students, faculty, and industry professionals.